

Capital-Level Factor Neutrality with Projected Partial Rebalancing

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Abstract

Active managers often remove factor exposure from alpha signals before portfolio construction. This paper studies a portfolio-construction method that instead imposes neutrality directly on the capital weights that are actually traded. The practical issue is that downstream construction rules can change the exposures of the final portfolio even when the input signal has been residualised.

At each rebalance, an exposure matrix F_t defines the monitored neutral subspace $\ker F_t^\top$. The portfolio is written as $u_t = Q_t \theta_t$, where the columns of Q_t form an orthonormal basis of this subspace. Optimisation is carried out in neutral coordinates, so every feasible portfolio satisfies the factor-neutrality equations on the implemented capital vector. The same representation gives an equality-constrained Markowitz interpretation and finite-sample exposure-leakage bounds when the construction and monitoring exposure matrices differ.

A projected partial-rebalancing rule handles the dynamic case. Before trading toward the new neutral target, the previous holding is projected into the current neutral subspace; the portfolio is then moved part way toward the full neutral target. Since both endpoints are neutral under the current exposure matrix, the interpolation path remains neutral.

Walk-forward tests across six research universes compare ordinary alpha, residualised alpha, the full neutral target, and projected partial neutral-coordinate portfolios. The tests show that neutral-coordinate portfolios have exposure leakage at numerical precision, while projected partial variants reduce turnover relative to the full neutral target under fixed validation choices.

Keywords: portfolio construction; factor neutrality; turnover control; constrained Markowitz optimisation; transaction costs; walk-forward validation.

1 Introduction

Active portfolio management usually separates signal generation from portfolio construction. A signal model produces scores, expected-return proxies, or rankings; the construction layer converts those inputs into capital weights subject to risk, leverage, turnover, benchmark, liquidity, and exposure controls. This paper studies the construction problem: how to impose factor neutrality on the implemented capital vector, after downstream construction rules have been applied.

A common preliminary step is to residualise the alpha signal against style, sector, beta, volatility, or other characteristics before optimisation. That operation can clarify the forecasting input, but it does not by itself constrain the exposures of the final portfolio. For an active sleeve, overlay, or monitored mandate, the relevant audit object is the traded capital vector, not only the residualised signal. The problem is finite-sample and dynamic: the investable universe, exposure matrix, risk estimate, and neutral subspace change through time.

At rebalance time t , let $F_t \in \mathbb{R}^{n_t \times m_t}$ be the chosen exposure matrix. Its columns may contain sector indicators, style scores, beta estimates, volatility characteristics, nonlinear transformations of characteristics, or any other exposure directions to be removed. A capital vector $u_t \in \mathbb{R}^{n_t}$ is neutral when $F_t^\top u_t = 0$. Rather than residualising the signal and then optimising in the original asset

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coordinates, we compute an orthonormal basis Q_t of $\ker F_t^\top$ and optimise over portfolios of the form $u_t = Q_t \theta_t$. Thus the optimisation variables θ_t are coordinates inside the neutral capital space; every feasible coordinate vector automatically produces a neutral capital vector.

The paper develops a finite-sample construction layer that imposes neutrality on the implemented weights themselves. It also motivates the dynamic projection step: when the exposure matrix changes, the previous holding should first be projected into the current neutral subspace before turnover smoothing is applied.

The empirical analysis tests these construction claims using walk-forward research panels. The tests ask whether the proposed construction enforces the monitored exposure constraints after optimisation and rebalancing, and whether partial rebalancing can reduce turnover while preserving capital-level neutrality. The tests evaluate construction mechanics rather than alpha models.

2 Related literature and construction workflow

The mean–variance theory of [Markowitz \(1952\)](#) provides the canonical optimisation problem for portfolio choice. In practice, however, constraints, estimation error, and implementation rules can materially change the final weights; see, for example, [Michaud \(1989\)](#), [DeMiguel et al. \(2009\)](#), and [Kolm et al. \(2014\)](#). Covariance shrinkage and portfolio constraints are common ways to stabilise allocations in finite samples ([Ledoit and Wolf, 2004](#); [Jagannathan and Ma, 2003](#)), while transaction costs and rebalancing rules are central to practical portfolio implementation ([Lobo et al., 2007](#); [Boyd et al., 2017](#)).

The construction is also connected to active factor-control methods. In active portfolio management, signals are often residualised against sector, style, beta, volatility, liquidity, or other exposure variables before optimisation; see [Grinold and Kahn \(2000\)](#) and [Clarke et al. \(2002\)](#). Signal residualisation is a common preprocessing step, but the contribution here is to impose the equality constraint at the capital-allocation stage. The projected partial-rebalancing rule extends the same idea to the dynamic setting.

The method can be used as a factor-neutral construction module. The manager chooses the exposure matrix F_t , computes a basis Q_t for the corresponding neutral subspace, and uses this basis inside the portfolio-construction step. The relevant audit object is the implemented portfolio, not the raw forecast.

In an existing construction system, the module can be inserted after signal generation and before trade generation. At each rebalance, it computes the neutral target, projects the old holding into the current neutral subspace, selects the rebalancing coefficient, and records the resulting exposures and turnover. The guarantees apply only to the exposure columns included in F_t . Dynamic rebalancing is handled by the project-then-trade rule formalised in [Algorithm 1](#).

3 Neutral-coordinate theory

3.1 Finite-sample capital-level neutrality

Let n_t denote the number of assets available at time t and let m_t denote the number of exposure functions to be neutralised. The exposure matrix $F_t \in \mathbb{R}^{n_t \times m_t}$ is a finite-sample matrix chosen by the manager; its columns may contain sector indicators, style scores, beta estimates, liquidity variables, volatility characteristics, or nonlinear transformations of such quantities.

Let $\mathcal{N}_t := \ker F_t^\top$, and choose an orthonormal basis $Q_t \in \mathbb{R}^{n_t \times d_t}$ of $\mathcal{N}_t$, where $d_t = n_t - \text{rank}(F_t)$. A neutral-coordinate portfolio is a capital vector of the form

$$u_t = Q_t \theta_t, \quad \theta_t \in \mathbb{R}^{d_t}. \tag{1}$$

Thus θ_t parameterises the implemented capital weights inside the neutral subspace.

Lemma 1 (Finite-sample capital-level neutrality). *Every neutral-coordinate portfolio $u_t = Q_t \theta_t$ satisfies $F_t^\top u_t = 0$. Consequently, the portfolio is exactly neutral to every exposure column included in F_t . For the numerical statement, let ε_{svd} denote the absolute tolerance used to identify numerical null-space directions in the singular-value decomposition, and let $\|\cdot\|_{2 \rightarrow 2}$ denote the spectral operator norm. Moreover, suppose Q_t is computed from a singular value decomposition of F_t . If the retained numerical null-space vectors correspond to singular values at most ε_{svd} , then*

$$\|F_t^\top Q_t\|_{2 \rightarrow 2} \leq \varepsilon_{\text{svd}}. \quad (2)$$

Thus any computed neutral-coordinate portfolio satisfies $\|F_t^\top Q_t \theta_t\|_2 \leq \varepsilon_{\text{svd}} \|\theta_t\|_2$.

Proof. Since the columns of Q_t lie in $\ker F_t^\top$, one has $F_t^\top Q_t = 0$. Thus, for every θ_t , $F_t^\top u_t = F_t^\top Q_t \theta_t = 0$. For the numerical statement, if the retained null-space vectors correspond to singular values at most ε_{svd} , then $\|F_t^\top Q_t x\|_2 \leq \varepsilon_{\text{svd}}$ for every unit vector x . Thus $\|F_t^\top Q_t\|_{2 \rightarrow 2} \leq \varepsilon_{\text{svd}}$, and the leakage bound follows by applying this operator bound to θ_t . $\square$

3.2 Markowitz interpretation and closed-form solution

Let a_t be the alpha vector, let Σ_t be the covariance or risk matrix used in construction, let $\gamma > 0$ be a risk-aversion parameter, and let $\mathcal{C} \subseteq \mathbb{R}^{n_t}$ denote the portfolio constraint set. The ordinary active Markowitz optimiser solves

$$u_t^{\text{ord}} = \arg \max_{u \in \mathcal{C}} \left\{ a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u \right\}. \quad (3)$$

The neutral-coordinate optimiser imposes neutrality by construction. Define

$$\Theta_t := \{\theta \in \mathbb{R}^{d_t} : Q_t \theta \in \mathcal{C}\}. \quad (4)$$

Then the constrained neutral-coordinate problem is

$$\theta_t^* = \arg \max_{\theta \in \Theta_t} \left\{ a_t^\top Q_t \theta - \frac{\gamma}{2} \theta^\top Q_t^\top \Sigma_t Q_t \theta \right\}, \quad u_t^{\text{NC}} = Q_t \theta_t^*. \quad (5)$$

Proposition 1 (Neutral constrained optimality). *The neutral-coordinate optimiser u_t^{NC} is exactly the Markowitz optimiser restricted to the neutral feasible set:*

$$u_t^{\text{NC}} = \arg \max_{u \in \mathcal{C} \cap \ker F_t^\top} \left\{ a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u \right\}. \quad (6)$$

Proof. The map $\theta \mapsto Q_t \theta$ is a one-to-one parametrisation of $\ker F_t^\top$, with inverse $u \mapsto Q_t^\top u$ on that subspace. The constraint $Q_t \theta \in \mathcal{C}$ is exactly the condition $u \in \mathcal{C} \cap \ker F_t^\top$. Substituting $u = Q_t \theta$ in the Markowitz objective gives the stated coordinate problem. $\square$

Remark 1 (Basis invariance). *The feasible capital set does not depend on the particular orthonormal basis used for $\ker F_t^\top$. If Q_t and $\tilde{Q}_t$ are two orthonormal bases of the same neutral subspace, then*

$$\{Q_t \theta : \theta \in \mathbb{R}^{d_t}\} = \{\tilde{Q}_t \eta : \eta \in \mathbb{R}^{d_t}\} = \ker F_t^\top.$$

Thus the coordinates may change under a different SVD basis, but the feasible capital portfolios and any unique capital-space optimiser are unchanged.

The neutral-coordinate construction is equivalent to the equality-constrained Markowitz problem. For this subsection, ignore box, leverage, and gross-exposure constraints and consider only the finite-sample neutrality constraint:

$$\max_{u \in \mathbb{R}^{n_t}} \left\{ a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u \right\} \quad \text{subject to} \quad F_t^\top u = 0. \quad (7)$$

Proposition 2 (Closed-form neutral Markowitz solution). *Assume $\Sigma_t \succ 0$ and $F_t^\top \Sigma_t^{-1} F_t$ is nonsingular. Then the equality-constrained optimiser is*

$$u_t^{\text{neu}} = \frac{1}{\gamma} P_t^\Sigma \Sigma_t^{-1} a_t, \quad (8)$$

where

$$P_t^\Sigma := I - \Sigma_t^{-1} F_t (F_t^\top \Sigma_t^{-1} F_t)^{-1} F_t^\top \quad (9)$$

is the Σ_t -metric projection onto $\ker F_t^\top$. If Q_t is an orthonormal basis of $\ker F_t^\top$, the same optimiser is obtained in neutral coordinates:

$$\theta_t^* = \frac{1}{\gamma} (Q_t^\top \Sigma_t Q_t)^{-1} Q_t^\top a_t, \quad u_t^{\text{neu}} = Q_t \theta_t^*. \quad (10)$$

This formula is the usual unconstrained Markowitz allocation with the component that would create exposure to F_t removed in the Σ_t -metric. The neutral-coordinate formula gives the same portfolio by optimising directly inside the neutral subspace.

Proof. The Lagrangian is

$$\mathcal{L}(u, \lambda) = a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u - \lambda^\top F_t^\top u.$$

The first-order condition with respect to u is $a_t - \gamma \Sigma_t u - F_t \lambda = 0$, so $u = \frac{1}{\gamma} \Sigma_t^{-1} (a_t - F_t \lambda)$. Imposing $F_t^\top u = 0$ gives $F_t^\top \Sigma_t^{-1} F_t \lambda = F_t^\top \Sigma_t^{-1} a_t$. Since $F_t^\top \Sigma_t^{-1} F_t$ is nonsingular,

$$\lambda = (F_t^\top \Sigma_t^{-1} F_t)^{-1} F_t^\top \Sigma_t^{-1} a_t.$$

Substituting this expression into the formula for u and collecting terms gives the stated P_t^Σ form.

For the coordinate form, write $u = Q_t \theta$. The objective becomes $a_t^\top Q_t \theta - \frac{\gamma}{2} \theta^\top Q_t^\top \Sigma_t Q_t \theta$. Since $\Sigma_t \succ 0$ and Q_t has full column rank, $Q_t^\top \Sigma_t Q_t \succ 0$. The first-order condition is $Q_t^\top a_t - \gamma Q_t^\top \Sigma_t Q_t \theta = 0$, so

$$\theta_t^* = \frac{1}{\gamma} (Q_t^\top \Sigma_t Q_t)^{-1} Q_t^\top a_t.$$

Because $Q_t \theta$ parametrises exactly $\ker F_t^\top$, this coordinate solution is the same constrained optimiser. $\square$

3.3 Relation to signal residualisation

A common alternative to neutral portfolio construction is to residualise the alpha signal against the exposure matrix. Neutral coordinates are different: they impose neutrality on the final capital allocation. The two operations coincide only under special risk geometry. Let $P_t := Q_t Q_t^\top$ be the Euclidean projector onto $\mathcal{N}_t = \ker F_t^\top$. The Euclidean residualised alpha is $a_t^{\text{res}} := P_t a_t$.

Proposition 3 (Residualisation agrees with neutral-coordinate only in special cases). *Assume there are no box, leverage, or gross-exposure constraints.*

1. If $\Sigma_t = \sigma_t^2 I$, then the neutral-coordinate optimiser is

$$u_t^{\text{NC}} = \frac{1}{\gamma \sigma_t^2} P_t a_t.$$

Thus, under isotropic risk, neutral-coordinate optimisation is proportional to the residualised alpha signal.

2. For general $\Sigma_t \succ 0$, the neutral-coordinate optimiser is

$$u_t^{\text{NC}} = \frac{1}{\gamma} Q_t (Q_t^\top \Sigma_t Q_t)^{-1} Q_t^\top a_t.$$

In general this is not proportional to $P_t a_t$. Thus, residualising the alpha signal is not equivalent to imposing neutrality on the final capital allocation.

Proof. If $\Sigma_t = \sigma_t^2 I$, then $Q_t^\top \Sigma_t Q_t = \sigma_t^2 Q_t^\top Q_t = \sigma_t^2 I$. The coordinate solution from Proposition 2 gives $u_t^{\text{NC}} = Q_t (\gamma \sigma_t^2)^{-1} Q_t^\top a_t = (\gamma \sigma_t^2)^{-1} P_t a_t$. For general Σ_t , the restricted risk matrix $Q_t^\top \Sigma_t Q_t$ need not be a scalar multiple of the identity. The optimiser depends on the inverse of this restricted risk matrix, which generally differs from the Euclidean projection $Q_t Q_t^\top a_t$. The two agree only when the risk matrix acts isotropically on the neutral subspace, or in other special cases where the restricted inverse reduces to a scalar multiple of the identity. $\square$

Proposition 4 (Risk-coupling form of residualised-alpha leakage). *Assume $\Sigma_t \succ 0$. Let Q_t be an orthonormal basis of $\mathcal{N}_t = \ker F_t^\top$, and let $Q_t^\perp$ be an orthonormal basis of $\mathcal{N}_t^\perp = \text{range}(F_t)$. In the orthonormal basis $(Q_t \ Q_t^\perp)$, write*

$$\begin{pmatrix} Q_t^\top \\ (Q_t^\perp)^\top \end{pmatrix} \Sigma_t \begin{pmatrix} Q_t & Q_t^\perp \end{pmatrix} = \begin{pmatrix} A_t & B_t \\ B_t^\top & D_t \end{pmatrix}.$$

Let $S_t := A_t - B_t D_t^{-1} B_t^\top$ be the Schur complement of D_t . For an alpha vector a_t , set $x_t := Q_t^\top a_t$ and residualise the alpha by $a_t^{\text{res}} = Q_t Q_t^\top a_t = Q_t x_t$. The unconstrained optimiser using the residualised alpha is

$$u_t^{\text{res}} = \frac{1}{\gamma} \Sigma_t^{-1} a_t^{\text{res}}.$$

Then

$$u_t^{\text{res}} = \frac{1}{\gamma} \left[Q_t S_t^{-1} x_t - Q_t^\perp D_t^{-1} B_t^\top S_t^{-1} x_t \right],$$

and consequently

$$F_t^\top u_t^{\text{res}} = -\frac{1}{\gamma} F_t^\top Q_t^\perp D_t^{-1} B_t^\top S_t^{-1} x_t.$$

In particular, residualising the alpha signal produces neutral capital for every a_t if and only if $B_t = 0$. Equivalently, the neutral subspace $\mathcal{N}_t$ is invariant under Σ_t , or equivalently under Σ_t^{-1} .

Moreover, if

$$u_t^{\text{NC}} = \frac{1}{\gamma} Q_t A_t^{-1} x_t$$

denotes the neutral-coordinate Markowitz optimiser for the same alpha coordinates, then

$$u_t^{\text{res}} - u_t^{\text{NC}} = \frac{1}{\gamma} \left[Q_t (S_t^{-1} - A_t^{-1}) x_t - Q_t^\perp D_t^{-1} B_t^\top S_t^{-1} x_t \right],$$

and hence

$$\|u_t^{\text{res}} - u_t^{\text{NC}}\|_2 \leq \frac{\|x_t\|_2}{\gamma} \left(\|S_t^{-1} B_t D_t^{-1} B_t^\top A_t^{-1}\|_{2 \rightarrow 2} + \|D_t^{-1} B_t^\top S_t^{-1}\|_{2 \rightarrow 2} \right).$$

Thus the leakage and the difference from the neutral-coordinate optimiser are controlled by the risk-coupling block B_t between the neutral subspace and its exposure complement.

Proof. In the basis $(Q_t \ Q_t^\perp)$, the residualised alpha has coordinates $(x_t^\top \ 0)^\top$. The inverse of the block matrix

$$\begin{pmatrix} A_t & B_t \\ B_t^\top & D_t \end{pmatrix}$$

has first column block

$$\begin{pmatrix} S_t^{-1} \\ -D_t^{-1}B_t^\top S_t^{-1} \end{pmatrix}, \quad S_t = A_t - B_t D_t^{-1} B_t^\top.$$

Therefore

$$\Sigma_t^{-1} Q_t x_t = Q_t S_t^{-1} x_t - Q_t^\perp D_t^{-1} B_t^\top S_t^{-1} x_t,$$

which gives the displayed formula for u_t^{res} . Applying $F_t^\top$ gives the leakage identity, since $F_t^\top Q_t = 0$.

Because D_t and S_t are positive definite, the leakage vanishes for every x_t if and only if $B_t = 0$. This is precisely the condition that Σ_t has no off-diagonal block coupling between $\mathcal{N}_t$ and $\mathcal{N}_t^\perp$. Equivalently, Σ_t maps $\mathcal{N}_t$ into itself. Since Σ_t is positive definite and symmetric, this is also equivalent to Σ_t^{-1} mapping $\mathcal{N}_t$ into itself.

The neutral-coordinate optimiser is

$$u_t^{\text{NC}} = \frac{1}{\gamma} Q_t (Q_t^\top \Sigma_t Q_t)^{-1} Q_t^\top a_t = \frac{1}{\gamma} Q_t A_t^{-1} x_t.$$

Subtracting this from the formula for u_t^{res} gives the displayed difference identity. Finally,

$$S_t^{-1} - A_t^{-1} = S_t^{-1} (A_t - S_t) A_t^{-1} = S_t^{-1} B_t D_t^{-1} B_t^\top A_t^{-1},$$

and the stated norm bound follows from the triangle inequality and submultiplicativity of the operator norm. $\square$

Remark 2 (Interpretation). *The block $B_t = Q_t^\top \Sigma_t Q_t^\perp$ measures how the construction risk model couples neutral directions to exposure directions. If $B_t = 0$, residualising the alpha signal and then applying the unconstrained optimiser is harmless: the optimiser cannot move capital out of the neutral subspace. If $B_t \neq 0$, the inverse risk map mixes neutral alpha coordinates into exposure directions, and the implemented capital vector generally leaks exposure. The size of this leakage is governed by*

$$\|D_t^{-1} B_t^\top S_t^{-1}\|_{2 \rightarrow 2}.$$

Thus residualisation fails not because the alpha was insufficiently projected, but because the risk model rotates the residualised signal out of the neutral capital subspace during optimisation.

The neutrality guarantee is exposure-relative: it applies to the exposure matrix F_t used in the construction. If the monitored or realised exposure matrix is instead $\tilde{F}_t = F_t + \Delta_t$, then a neutral-coordinate portfolio satisfies $\tilde{F}_t^\top u_t = \Delta_t^\top u_t$, so the remaining leakage is controlled by the exposure-definition error. The corresponding finite-sample bound and numerical null-space error bound are recorded in Appendix A.

4 Projected partial neutral-coordinate rebalancing

4.1 Projected partial rebalancing and line search

The neutral subspace may change from one rebalance date to the next. The rule below controls turnover directly in capital space by projecting the previous holding into the current neutral subspace and then trading along a neutral line segment.

Let $P_{\mathcal{N}_t} = Q_t Q_t^\top$ be the Euclidean projector onto $\mathcal{N}_t$. First project yesterday's held portfolio into the current neutral subspace:

$$u_{t-1 \rightarrow t} := P_{\mathcal{N}_t} u_{t-1}. \quad (11)$$

Then compute the full neutral target $u_t^* \in \mathcal{N}_t$ and trade only part of the way:

$$u_t(\beta) = (1 - \beta)u_{t-1 \rightarrow t} + \beta u_t^*, \quad \beta \in [0, 1]. \quad (12)$$

The coefficient β controls trading speed: $\beta = 0$ keeps the projected old holding, $\beta = 1$ moves fully to the new neutral target, and intermediate values trade partway while remaining neutral. The implemented rule chooses β by the one-dimensional utility problem

$$\max_{\beta \in [0, 1]} \left\{ a_t^\top u_t(\beta) - \frac{\gamma}{2} u_t(\beta)^\top \Sigma_t u_t(\beta) - c \|u_t(\beta) - u_{t-1}\|_1 \right\}. \quad (13)$$

Proposition 5 (Projected partial neutral-coordinate rebalancing). *Let Q_t be an orthonormal basis of $\mathcal{N}_t = \ker F_t^\top$, and let $P_{\mathcal{N}_t} = Q_t Q_t^\top$ be the Euclidean projector onto the current neutral subspace. Given the previously held portfolio u_{t-1} , define its current neutral projection by*

$$u_{t-1 \rightarrow t} := P_{\mathcal{N}_t} u_{t-1}.$$

Let $u_t^* \in \mathcal{N}_t$ be the full neutral target, set $d_t := u_t^* - u_{t-1 \rightarrow t}$, and define $u_t(\beta) := u_{t-1 \rightarrow t} + \beta d_t$ for $\beta \in [0, 1]$. Then:

1. For every $\beta \in [0, 1]$, $F_t^\top u_t(\beta) = 0$.
2. The distance from the projected previous holding to the full neutral target satisfies

$$\|u_t(\beta) - u_{t-1 \rightarrow t}\|_1 = \beta \|u_t^* - u_{t-1 \rightarrow t}\|_1. \quad (14)$$

3. The actual trade from the previously held portfolio satisfies

$$\|u_t(\beta) - u_{t-1}\|_1 \leq \|u_{t-1 \rightarrow t} - u_{t-1}\|_1 + \beta \|u_t^* - u_{t-1 \rightarrow t}\|_1. \quad (15)$$

4. Suppose $\Sigma_t \succeq 0$, $\gamma > 0$, $c \geq 0$, and $d_t^\top \Sigma_t d_t > 0$. Then the one-dimensional turnover-penalised objective

$$\Phi_t(\beta) = a_t^\top u_t(\beta) - \frac{\gamma}{2} u_t(\beta)^\top \Sigma_t u_t(\beta) - c \|u_t(\beta) - u_{t-1}\|_1 \quad (16)$$

is concave and piecewise quadratic on $[0, 1]$. Its global maximiser is obtained by checking the endpoints, all sign-change breakpoints

$$\mathcal{B}_t := \{\beta \in (0, 1) : u_{t-1 \rightarrow t, i} + \beta d_{t, i} - u_{t-1, i} = 0 \text{ for some } i \text{ with } d_{t, i} \neq 0\},$$

and the interval-wise critical points between consecutive breakpoints.

Proof. For neutrality, note that $u_{t-1 \rightarrow t} = P_{\mathcal{N}_t} u_{t-1} \in \mathcal{N}_t$ by definition of the orthogonal projector, while $u_t^* \in \mathcal{N}_t$ by assumption. Thus $d_t = u_t^* - u_{t-1 \rightarrow t} \in \mathcal{N}_t$, because $\mathcal{N}_t$ is a linear subspace, and $u_t(\beta) = u_{t-1 \rightarrow t} + \beta d_t \in \mathcal{N}_t$ for every $\beta \in [0, 1]$. Since $\mathcal{N}_t = \ker F_t^\top$, this proves $F_t^\top u_t(\beta) = 0$.

The distance identity follows from positive homogeneity of the ℓ_1 -norm applied to $u_t(\beta) - u_{t-1 \rightarrow t} = \beta(u_t^* - u_{t-1 \rightarrow t})$. The actual-trade bound follows from the triangle inequality applied to $u_t(\beta) - u_{t-1} = (u_{t-1 \rightarrow t} - u_{t-1}) + \beta(u_t^* - u_{t-1 \rightarrow t})$.

It remains to prove the line-search statement. Write $u_0 = u_{t-1 \rightarrow t}$ and $d = d_t$. The Markowitz part of the objective is

$$a_t^\top (u_0 + \beta d) - \frac{\gamma}{2} (u_0 + \beta d)^\top \Sigma_t (u_0 + \beta d).$$

Expanding in β , this equals

$$\text{constant} + \beta \left(a_t^\top d - \gamma d^\top \Sigma_t u_0 \right) - \frac{\gamma}{2} \beta^2 d^\top \Sigma_t d.$$

Since $d^\top \Sigma_t d > 0$, this is strictly concave quadratic in β . The turnover term is

$$\|u_0 + \beta d - u_{t-1}\|_1 = \sum_i |u_{0,i} + \beta d_i - u_{t-1,i}|.$$

Each summand is convex and piecewise linear in β , with a possible kink only at $u_{0,i} + \beta d_i - u_{t-1,i} = 0$. Thus the negative turnover penalty is concave and piecewise linear, and Φ_t is concave and piecewise quadratic. On any open interval between consecutive breakpoints, the sign vector $s_i = \text{sign}(u_{0,i} + \beta d_i - u_{t-1,i})$ is constant. On such an interval,

$$\|u_0 + \beta d - u_{t-1}\|_1 = s^\top (u_0 + \beta d - u_{t-1}),$$

so

$$\Phi'_t(\beta) = a_t^\top d - \gamma d^\top \Sigma_t u_0 - \gamma \beta d^\top \Sigma_t d - c s^\top d.$$

The unique critical point on that interval, if it lies in the interval, is thus

$$\beta_s^* = \frac{a_t^\top d - \gamma d^\top \Sigma_t u_0 - c s^\top d}{\gamma d^\top \Sigma_t d}.$$

A concave piecewise quadratic function on a compact interval attains its maximum at either an endpoint, a breakpoint, or an interval-wise critical point. Checking this finite set thus gives a global maximiser. $\square$

Remark 3 (Why the projection is needed). *If one partially rebalances directly from the old holding,*

$$\tilde{u}_t(\beta) = (1 - \beta)u_{t-1} + \beta u_t^*, \quad 0 \leq \beta \leq 1,$$

then, since $F_t^\top u_t^ = 0$,*

$$F_t^\top \tilde{u}_t(\beta) = (1 - \beta)F_t^\top u_{t-1}.$$

Thus the unprojected path is neutral under the current exposure matrix only if $u_{t-1} \in \ker F_t^\top$ or $\beta = 1$. The projection step repairs this before turnover smoothing is applied.

Proposition 6 (Forced turnover from neutral-subspace rotation). *Let*

$$\mathcal{N}_{t-1} = \ker F_{t-1}^\top, \quad \mathcal{N}_t = \ker F_t^\top,$$

and let P_{t-1} and P_t be the corresponding Euclidean orthogonal projectors. Suppose the previous holding is neutral at time $t - 1$, so $u_{t-1} \in \mathcal{N}_{t-1}$. Define the projection-repair trade

$$r_t := P_t u_{t-1} - u_{t-1}.$$

Then

$$\|r_t\|_2 = \text{dist}_2(u_{t-1}, \mathcal{N}_t) = \|(I - P_t)u_{t-1}\|_2.$$

Consequently, every portfolio $v \in \mathcal{N}_t$ satisfies $\|v - u_{t-1}\|_2 \geq \|r_t\|_2$. Thus $\|r_t\|_2$ is the unavoidable Euclidean trade required to restore neutrality under the current exposure matrix.

Moreover,

$$\|r_t\|_2 = \|(I - P_t)P_{t-1}u_{t-1}\|_2 \leq \|(I - P_t)P_{t-1}\|_{2 \rightarrow 2} \|u_{t-1}\|_2.$$

If $\mathcal{N}_{t-1}$ and $\mathcal{N}_t$ have the same dimension, then

$$\|(I - P_t)P_{t-1}\|_{2 \rightarrow 2} = \sin \theta_{\max}(\mathcal{N}_{t-1}, \mathcal{N}_t),$$

where $\theta_{\max}(\mathcal{N}_{t-1}, \mathcal{N}_t)$ is the largest principal angle between the two neutral subspaces. Therefore

$$\|r_t\|_2 \leq \sin \theta_{\max}(\mathcal{N}_{t-1}, \mathcal{N}_t) \|u_{t-1}\|_2.$$

For the projected partial rule

$$u_t(\beta) = P_t u_{t-1} + \beta(u_t^* - P_t u_{t-1}), \quad 0 \leq \beta \leq 1,$$

one also has $\|u_t(\beta) - u_{t-1}\|_2 \geq \|r_t\|_2$. Thus the projection step creates a turnover floor determined by the rotation of the neutral subspace, before any discretionary movement toward the new target is made.

Proof. Since $P_t u_{t-1}$ is the Euclidean orthogonal projection of u_{t-1} onto the closed linear subspace $\mathcal{N}_t$, the projection theorem gives

$$\|u_{t-1} - P_t u_{t-1}\|_2 = \inf_{v \in \mathcal{N}_t} \|u_{t-1} - v\|_2 = \text{dist}_2(u_{t-1}, \mathcal{N}_t).$$

This proves the first identity and the lower bound for every $v \in \mathcal{N}_t$.

Because $u_{t-1} \in \mathcal{N}_{t-1}$, one has $u_{t-1} = P_{t-1} u_{t-1}$. Hence

$$r_t = P_t u_{t-1} - u_{t-1} = -(I - P_t)P_{t-1} u_{t-1},$$

and therefore

$$\|r_t\|_2 \leq \|(I - P_t)P_{t-1}\|_{2 \rightarrow 2} \|u_{t-1}\|_2.$$

When the two subspaces have the same dimension, the operator norm $\|(I - P_t)P_{t-1}\|_{2 \rightarrow 2}$ equals the sine of the largest principal angle between them. This gives the principal-angle bound.

Finally, $u_t(\beta) \in \mathcal{N}_t$ for every $\beta \in [0, 1]$ by Proposition 5. Applying the already proved lower bound with $v = u_t(\beta)$ gives

$$\|u_t(\beta) - u_{t-1}\|_2 \geq \text{dist}_2(u_{t-1}, \mathcal{N}_t) = \|r_t\|_2.$$

□

Remark 4 (Interpretation). *The projected partial rule separates turnover into two conceptually different pieces. The first piece, $P_t u_{t-1} - u_{t-1}$, is not a trade toward a new alpha target; it is the minimum Euclidean repair needed because the monitored neutral subspace has changed. The second piece, $\beta(u_t^* - P_t u_{t-1})$, is the discretionary trade toward the new neutral Markowitz target. Thus fast rotation of F_t can create turnover even when the strategy chooses $\beta = 0$. This gives a direct diagnostic for involuntary turnover: monitor*

$$\|(I - P_t)P_{t-1}\|_{2 \rightarrow 2}$$

or, when dimensions match, the largest principal angle between consecutive neutral subspaces.

The full projected-partial construction used in the validation is summarised in Algorithm 1.

Algorithm 1. Projected partial neutral-coordinate rebalancing

1. Construct the exposure matrix F_t from information available before the holding interval.
2. Compute an orthonormal basis Q_t of $\mathcal{N}_t = \ker F_t^\top$.
3. Compute the full neutral target $u_t^* = Q_t \theta_t^*$ by solving the neutral Markowitz problem in neutral coordinates.
4. Project the previous holding into the current neutral subspace: $u_{t-1 \rightarrow t} = Q_t Q_t^\top u_{t-1}$.
5. Set $d_t = u_t^* - u_{t-1 \rightarrow t}$.
6. Choose $\beta_t \in [0, 1]$ to maximise

$$a_t^\top (u_{t-1 \rightarrow t} + \beta d_t) - \frac{\gamma}{2} (u_{t-1 \rightarrow t} + \beta d_t)^\top \Sigma_t (u_{t-1 \rightarrow t} + \beta d_t) - c \|u_{t-1 \rightarrow t} + \beta d_t - u_{t-1}\|_1.$$

7. Hold $u_t = (1 - \beta_t)u_{t-1 \rightarrow t} + \beta_t u_t^*$.
8. Record neutrality leakage $\|F_t^\top u_t\|_\infty$, turnover $\|u_t - u_{t-1}\|_1$, ex-ante risk, selected β_t , and realised holding-period return.

When the turnover penalty is omitted, the one-dimensional line search has a closed form. Let

$$u_0 := u_{t-1 \rightarrow t}, \quad d := u_t^* - u_0, \quad u_t(\beta) = u_0 + \beta d.$$

Define

$$\phi_0(\beta) = a_t^\top (u_0 + \beta d) - \frac{\gamma}{2} (u_0 + \beta d)^\top \Sigma_t (u_0 + \beta d). \quad (17)$$

Then

$$\phi_0(\beta) = \phi_0(0) + \beta \left(a_t^\top d - \gamma d^\top \Sigma_t u_0 \right) - \frac{\gamma}{2} \beta^2 d^\top \Sigma_t d. \quad (18)$$

Lemma 2 (Closed-form no-cost partial-rebalance fraction). *Assume $\Sigma_t \succeq 0$, $\gamma > 0$, and $d^\top \Sigma_t d > 0$. Without the turnover penalty, the optimal coefficient on $[0, 1]$ is*

$$\beta_0^* = \Pi_{[0,1]} \left(\frac{a_t^\top d - \gamma d^\top \Sigma_t u_0}{\gamma d^\top \Sigma_t d} \right), \quad (19)$$

where $\Pi_{[0,1]}(x) = \min\{1, \max\{0, x\}\}$.

Proof. Differentiating the displayed quadratic gives

$$\phi_0'(\beta) = a_t^\top d - \gamma d^\top \Sigma_t u_0 - \gamma \beta d^\top \Sigma_t d.$$

Since $d^\top \Sigma_t d > 0$, the function is strictly concave in β . The unconstrained maximiser is the displayed fraction, and projecting this value onto $[0, 1]$ gives the constrained maximiser. $\square$

4.2 Implementation overlays and projection repair

Scalar overlays preserve neutrality. If $u_t \in \ker F_t^\top$ and $s_t \in \mathbb{R}$, then $F_t^\top(s_t u_t) = s_t F_t^\top u_t = 0$. Thus scalar volatility targeting, scalar gross-exposure normalisation, and scalar risk-budget adjustment do not introduce exposure leakage.

In contrast, assetwise non-scalar overlays need not preserve neutrality. If an implementation overlay acts through a diagonal matrix $D_t = \text{diag}(d_{1,t}, \dots, d_{n_t,t})$, then $F_t^\top D_t u_t$ is generally nonzero even when $F_t^\top u_t = 0$. Such operations should be handled inside the constrained optimiser or followed by a projection back into $\ker F_t^\top$.

Lemma 3 (Projection repair after an implementation overlay). *Let $v_t \in \mathbb{R}^{n_t}$ be any portfolio obtained after an arbitrary implementation operation, such as clipping, assetwise scaling, or external trade adjustment. Define*

$$\Pi_t v_t := Q_t Q_t^\top v_t.$$

Then $\Pi_t v_t \in \ker F_t^\top$. Moreover, among all neutral portfolios $u \in \ker F_t^\top$, $\Pi_t v_t$ is the unique Euclidean closest portfolio to v_t :

$$\Pi_t v_t = \arg \min_{u \in \ker F_t^\top} \|u - v_t\|_2.$$

Proof. Since $Q_t Q_t^\top$ is the orthogonal projector onto $\mathcal{N}_t = \ker F_t^\top$, its range is exactly $\mathcal{N}_t$. Hence $\Pi_t v_t \in \mathcal{N}_t$ and $F_t^\top \Pi_t v_t = 0$. The closest-point statement is the standard Euclidean projection theorem for closed linear subspaces. $\square$

5 Experimental design

5.1 Research universes and baselines

The experiment suite uses six research panels: US large-cap sample, Sector, Style, GlobalCountries, USBonds, and Country. The label “US large-cap sample” refers to a 48-asset cleaned level panel used in the local research files; it is not a historical S&P 500 membership file. The empirical comparison holds the alpha inputs and exposure construction fixed across construction rules. The core baselines are:

1. ordinary alpha, which uses the raw signal in the ordinary portfolio-construction layer;
2. residualised alpha, which first residualises the signal against F_t but does not enforce capital-space neutrality after construction;
3. full neutral-coordinate target, equivalently the full constrained neutral Markowitz target, which moves directly to the exact neutral target with no turnover-smoothing step;
4. projected partial neutral-coordinate portfolio, which projects the previously held portfolio into $\mathcal{N}_t$ and then partially rebalances according to (13).

The three projected-partial candidates reported below are:

- Defensive momentum neutral-coordinate: a low-turnover candidate with β capped at 0.25;
- Ensemble continuous neutral-coordinate: the balanced empirical candidate using a signal ensemble and continuous β ;
- Reversal continuous neutral-coordinate: a reversal-based comparison candidate with continuous β .

The candidates are fixed before the reported universe-level comparison and are not retuned separately by universe.

Universe	Sample period	Daily observations	Rebalances	Assets
Country	2006-02-28–2025-11-28	4,970	237	10
GlobalCountries	2006-01-04–2025-12-11	5,017	239	16
US large-cap sample	2011-01-04–2025-12-03	3,752	179	48
Sector	2001-01-03–2025-12-11	6,346	303	11
Style	2006-02-28–2025-11-28	4,970	237	10
USBonds	2006-01-04–2025-12-11	5,017	239	13

Table 1: Research-panel coverage used in the fixed validation protocol. Asset counts are the assets with non-empty weight histories in the local validation files.

5.2 Validation metrics

The validation reports neutrality leakage, implementation quality, and stability diagnostics. Neutrality leakage is measured by $\|F_t^\top u_t\|_\infty$ at each rebalance and aggregated by the maximum observed value. Implementation quality is summarised by Sharpe ratio, median Sharpe across universes, turnover reduction versus the full neutral-coordinate target, and maximum-drawdown improvement. Stability checks include stationary-bootstrap comparisons (Politis and Romano, 1994), transaction-cost sensitivity, subperiod aggregates, and exposure-stress linearity.

Sharpe ratios are computed from the walk-forward portfolio return series using the same sampling convention across all strategies and universes. Turnover is measured at rebalance dates after gross normalisation. “TO red.” denotes turnover reduction relative to the full neutral-coordinate target, and “MaxDD imp.” denotes improvement in maximum drawdown relative to that target, so positive values indicate improvement.

The empirical comparison is not designed as a test of alpha discovery. All strategies use the same alpha inputs, rebalance dates, risk estimates, and exposure definitions. The empirical questions are whether neutral coordinates eliminate capital-level exposure leakage and whether projected partial rebalancing reduces turnover without leaving the neutral subspace.

6 Empirical results

6.1 Aggregate fixed validation

Table 2 compares the six research universes. Ordinary and residualised alpha separate signal-level from capital-level exposure control; the full neutral-coordinate portfolio is the constrained target, and projected partial variants trade part way toward the same target.

Strategy	Avg Sharpe	Med. Sharpe	Sharpe wins	TO red.	MaxDD imp.	Max exposure
Ordinary alpha	-0.189	-0.144	3	0.088	-0.103	14.312
Residualised alpha	-0.221	-0.194	0	-0.009	-0.017	2.785
Full neutral-coordinate target	-0.165	-0.144	0	0.000	0.000	4.83×10^{-15}
Defensive momentum neutral-coordinate	0.059	0.021	6	0.779	0.173	1.76×10^{-15}
Ensemble continuous neutral-coordinate	0.104	0.088	6	0.594	0.139	2.65×10^{-15}
Reversal continuous neutral-coordinate	0.194	0.193	5	0.230	0.124	4.97×10^{-15}

Table 2: Aggregate benchmark comparison across six research universes. Sharpe wins, turnover reduction, and maximum-drawdown improvement are measured relative to the full neutral-coordinate target.

The implemented neutral-coordinate portfolios satisfy the monitored neutrality constraints to numerical precision; detailed leakage audits are reported in Appendix A. Figure 1 summarises the turnover–Sharpe tradeoff for the projected partial variants; per-universe rows are reported in Appendix A, Table 6.

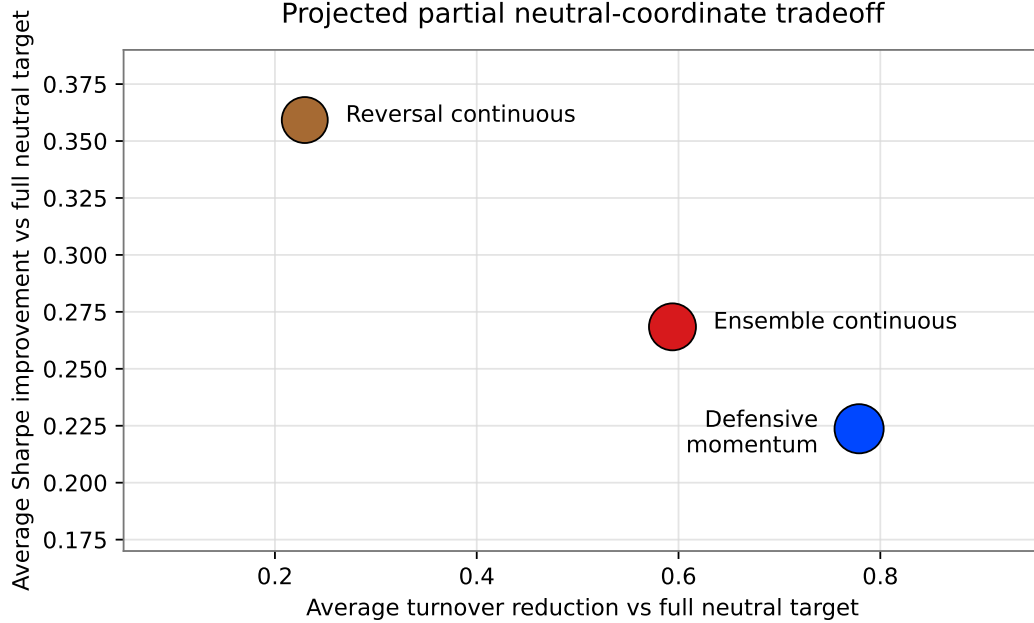


Figure 1: Projected partial neutral-coordinate tradeoff relative to the full neutral target.

6.2 Rebalancing coefficients and neutral-coordinate conditioning

Table 3 reports the partial-rebalance coefficient summaries for the three fixed projected-partial candidates. The defensive candidate has a binding cap by design, while the continuous candidates retain a substantial fraction of interior choices. Additional rank and conditioning checks for the neutral-coordinate basis are reported in Appendix A.

Strategy	Mean β	Median β	$\Pr(\beta = 0)$	$\Pr(\beta = 1)$	$\Pr(0 < \beta < 1)$	Turnover
Defensive momentum	0.175	0.250	0.038	0.000	0.962	0.196
Ensemble continuous	0.315	0.120	0.097	0.183	0.720	0.362
Reversal continuous	0.465	0.230	0.095	0.379	0.526	0.682

Table 3: Projected partial rebalancing coefficients averaged across the six universes.

6.3 Robustness and sensitivity checks

The fixed validation uses the cubic volatility specification $F_t = [1, \sigma, \sigma^2, \sigma^3]$. In the reported order sweep, the construction remains neutral when the exposure matrix is varied from $[1, \sigma]$ through $[1, \dots, \sigma^4]$, with zero rank failures. Appendix A, Table 7 reports exposure-stress checks. The maximum bound ratio remains below one and the minimum linearity R^2 is 0.970, consistent with Proposition 7.

Appendix A, Table 8 reports stationary-bootstrap uncertainty. Turnover reductions are stable across all six universes for the defensive and ensemble candidates; Sharpe and drawdown comparisons are positive in aggregate but less uniform across universes. Figure 2 reports sensitivity to realised transaction-cost assumptions. As realised costs rise, the lower-turnover projected partial portfolios improve relative to the full neutral target. Subperiod checks show similar turnover reductions.

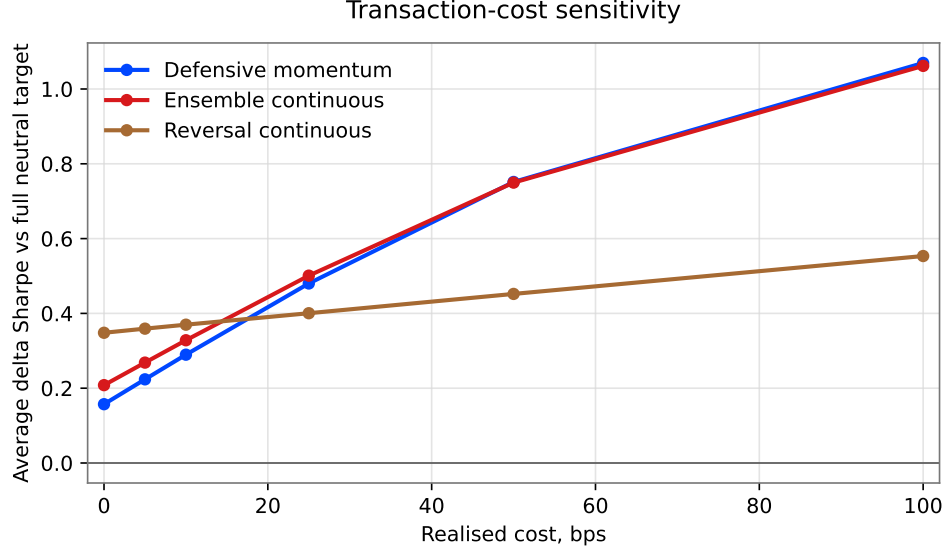


Figure 2: Transaction-cost sensitivity relative to the full neutral-coordinate target.

7 Research data audit and implementation diagnostics

The empirical panels are processed validation panels used to evaluate the portfolio-construction layer. Table 4 summarises the data-audit scope. Neutrality checks are computed directly from the implemented weights and recorded exposure matrices at each rebalance. Country and Style have local OHLCV proxy coverage with volume and spread-proxy fields, GlobalCountries has partial OHLCV proxy coverage, and the US large-cap sample, Sector, and USBonds are cleaned level-only panels.

Universe	Source type	Volume/spread proxy	Diagnostics used
Country	local proxy OHLCV panel	yes	construction + cost-accounting check
GlobalCountries	local proxy OHLCV panel	partial	construction + partial cost-accounting check
US large-cap sample	cleaned level panel	no	construction and neutrality checks
Sector	cleaned level panel	no	construction and neutrality checks
Style	local proxy OHLCV panel	yes	construction + cost-accounting check
USBonds	cleaned level panel	no	construction and neutrality checks

Table 4: Research data-audit scope for the processed validation panels.

The monitor records neutrality, selected beta, turnover, ex-ante risk, condition number, drawdown, weights, and trade lists at every rebalance.

Proxy cost-accounting checks were computed only for panels with local OHLCV proxy fields, including partial coverage for GlobalCountries. The detailed rows are included with the replication supplement rather than used as a main manuscript result.

8 Discussion and scope

The empirical results support the finite-sample neutrality claim and show lower turnover for the projected partial variants. The guarantees are finite-sample and exposure-relative. Given the monitored exposure matrix F_t , the construction enforces neutrality with respect to the selected exposure columns on the implemented capital vector. The empirical study uses fixed research panels, simple signal definitions, trailing covariance estimates, and a parsimonious volatility-exposure specification to keep the validation focused on the construction layer. A production implementation can change the alpha model, exposure definition, liquidity constraints, or transaction-cost model without changing the

neutral-coordinate construction.

9 Conclusion

This paper developed a finite-sample method for imposing factor neutrality on implemented portfolio weights rather than on the alpha signal alone. The neutral-coordinate representation is equivalent to equality-constrained Markowitz optimisation, while the projected partial rule preserves neutrality during turnover smoothing when the exposure matrix changes over time.

In the reported walk-forward tests, neutral-coordinate portfolios have exposure leakage at numerical precision, and projected partial variants reduce turnover relative to the full neutral target. The method gives an ex-post checkable implementation of factor-neutral active mandates at the level of traded capital weights.

Conflicts of Interest

The author declares no conflicts of interest.

Data and code availability

Replication code and scripts are available on reasonable request. Processed research panels and intermediate inputs are provided where redistribution is permitted. The empirical inputs are constructed from publicly available data sources, subject to the terms of the corresponding data providers.

A Additional technical and audit checks

This appendix records the fixed implementation protocol, per-universe results, exposure-stress checks, stationary-bootstrap uncertainty, and rank-conditioning diagnostics. Additional replication files include cost-accounting details, subperiod summaries, exposure-order sweeps, monitor logs, and proxy cost-accounting rows.

A.1 Exposure-misspecification and numerical leakage bounds

Lemma 4 (Exposure-misspecification leakage bound). *Let $u_t^{\text{NC}} \in \ker F_t^\top$, and let $\tilde{F}_t = F_t + \Delta_t$. Then*

$$\|\tilde{F}_t^\top u_t^{\text{NC}}\|_2 \leq \|\Delta_t\|_{2 \rightarrow 2} \|u_t^{\text{NC}}\|_2. \quad (20)$$

If additionally $\|u_t^{\text{NC}}\|_1 \leq G$, then $\|\tilde{F}_t^\top u_t^{\text{NC}}\|_2 \leq G \|\Delta_t\|_{2 \rightarrow 2}$.

Proof. Since $F_t^\top u_t^{\text{NC}} = 0$, we have $\tilde{F}_t^\top u_t^{\text{NC}} = (F_t + \Delta_t)^\top u_t^{\text{NC}} = \Delta_t^\top u_t^{\text{NC}}$. The first inequality follows from the operator-norm bound, and the second from $\|u\|_2 \leq \|u\|_1$. $\square$

Proposition 7 (Combined numerical and exposure-misspecification leakage). *Let $Q_t \in \mathbb{R}^{n_t \times d_t}$ be a computed neutral-coordinate basis satisfying $\|F_t^\top Q_t\|_{2 \rightarrow 2} \leq \varepsilon_{\text{svd}}$. Let the realised exposure matrix be $\tilde{F}_t = F_t + \Delta_t$. For every computed neutral-coordinate portfolio $u_t = Q_t \theta_t$,*

$$\|\tilde{F}_t^\top u_t\|_2 \leq \varepsilon_{\text{svd}} \|\theta_t\|_2 + \|\Delta_t\|_{2 \rightarrow 2} \|u_t\|_2. \quad (21)$$

If, in addition, $\|u_t\|_1 \leq G$, then $\|\tilde{F}_t^\top u_t\|_2 \leq \varepsilon_{\text{svd}} \|\theta_t\|_2 + G \|\Delta_t\|_{2 \rightarrow 2}$.

Proof. Using $\tilde{F}_t = F_t + \Delta_t$ and $u_t = Q_t \theta_t$, $\tilde{F}_t^\top u_t = F_t^\top Q_t \theta_t + \Delta_t^\top u_t$. Taking Euclidean norms and applying the triangle inequality gives $\|\tilde{F}_t^\top u_t\|_2 \leq \|F_t^\top Q_t \theta_t\|_2 + \|\Delta_t^\top u_t\|_2$. The assumed computed null-space error gives $\|F_t^\top Q_t \theta_t\|_2 \leq \|F_t^\top Q_t\|_{2 \rightarrow 2} \|\theta_t\|_2 \leq \varepsilon_{\text{svd}} \|\theta_t\|_2$. Also $\|\Delta_t^\top u_t\|_2 \leq \|\Delta_t\|_{2 \rightarrow 2} \|u_t\|_2$. Combining the inequalities gives the first bound. If $\|u_t\|_1 \leq G$, then $\|u_t\|_2 \leq \|u_t\|_1 \leq G$, which gives the second bound. $\square$

The maximum monitored neutrality errors are 4.94×10^{-15} before scalar volatility targeting and 6.00×10^{-15} after scalar volatility targeting.

A.2 Fixed implementation protocol and diagnostic tables

Component	Fixed validation specification
Rebalance frequency	Approximately monthly, implemented as 21-trading-day walk-forward rebalances.
Lookback window	252 trailing trading days, using only information available before the holding interval.
Exposure matrix	$F_t = [1, \sigma_t, \sigma_t^2, \sigma_t^3]$ in the fixed validation; $\sigma_{i,t}$ is the 252-day trailing volatility computed before the holding interval.
Risk estimate	Trailing covariance matrix with ridge regularisation in the optimisation step.
Signals	63-day momentum, 252-day momentum skipping 21 days, 21-day reversal, EWMA trend, volatility-scaled momentum, and equal-weight ensemble signals.
Fixed candidates	Defensive momentum; ensemble continuous; reversal continuous NC.
Partial-rebalance rule	One-dimensional utility maximisation over $\beta \in [0, 1]$.
Turnover penalty	Common fixed ex-ante coefficient 7500; held fixed across universes and candidates, not selected by universe-level performance. The defensive candidate also caps β at 0.25.
Transaction-cost sensitivity	Reported at 0, 5, 10, 25, 50, and 100 basis points.
Universe-specific retuning	None in the fixed validation.

Table 5: Fixed implementation protocol for the final comparison.

Universe	Full target S	Mom S	Mom TO red.	Ens. S	Ens. TO red.	Rev. S
Country	-0.107	0.048	0.746	0.075	0.462	0.105
GlobalCountries	-0.490	-0.059	0.745	0.047	0.546	0.499
US large-cap	-0.181	-0.006	0.745	0.193	0.635	0.282
Sector	-0.237	0.117	0.804	0.159	0.661	0.321
Style	-0.067	-0.055	0.812	0.046	0.606	-0.060
USBonds	0.094	0.307	0.821	0.101	0.654	0.020

Table 6: Per-universe fixed neutral-coordinate comparison. “S” denotes Sharpe and “TO red.” denotes turnover reduction versus the full neutral-coordinate target.

Universe	Max bound ratio	Median bound ratio	Min linearity R^2	Stress pass rate
Country	0.664	0.093	0.989	1.000
GlobalCountries	0.579	0.081	0.983	1.000
US large-cap	0.460	0.047	0.970	1.000
Sector	0.837	0.222	0.994	1.000
Style	0.763	0.115	0.989	1.000
USBonds	0.796	0.304	0.986	1.000

Table 7: Exposure-stress linearity and leakage-bound checks.

Strategy	Metric	Avg delta	Win rate	CI excludes 0	N
Defensive momentum neutral-coordinate	Sharpe	0.224	0.869	2	6
Defensive momentum neutral-coordinate	Turnover	-0.689	1.000	6	6
Defensive momentum neutral-coordinate	Max drawdown	0.173	0.997	6	6
Ensemble continuous neutral-coordinate	Sharpe	0.268	0.861	3	6
Ensemble continuous neutral-coordinate	Turnover	-0.524	1.000	6	6
Ensemble continuous neutral-coordinate	Max drawdown	0.139	0.927	1	6
Reversal continuous neutral-coordinate	Sharpe	0.359	0.737	2	6
Reversal continuous neutral-coordinate	Turnover	-0.203	0.938	4	6
Reversal continuous neutral-coordinate	Max drawdown	0.124	0.812	1	6

Table 8: Stationary-bootstrap comparison against the full neutral-coordinate target. CI counts report the number of universes, out of six, where the 95 percent bootstrap interval excludes zero.

Universe	Median rank F_t	Min d_t	Median d_t	Median condition	Max leakage
Country	4	6	6	2.106	7.12×10^{-16}
GlobalCountries	4	12	12	3.563	1.54×10^{-15}
US large-cap	4	44	44	9.584	2.65×10^{-15}
Sector	4	7	7	1.028	4.81×10^{-16}
Style	4	6	6	2.032	6.48×10^{-16}
USBonds	4	9	9	6.019	9.07×10^{-16}

Table 9: Rank and conditioning checks for the main candidate monitor. Here $d_t = \dim \ker F_t^\top$.

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